Description:
The goal of this project is to explore the relationship between NBA players' in-game performance and their respective salaries. This will be gathered from data sources containing game statistics and salary information for each player.

Data Sources:

- Flat File:
  - Description: NBA player stats by season. Kaggle available datasets include games data, games details, player details, NBA rankings, and teams details.
  - Link: https://www.kaggle.com/datasets/nathanlauga/nba-games
- API:
  - Description: NBA real-time or historical player stats. Data includes team, player, and game points, assists, rebounds, etc.
  - Link: https://docs.balldontlie.io/#game-player-stats
- Website:
  - Description: Current contracts and salaries by player. Table includes player name, salary, and team.
  - Link: https://www.basketball-reference.com/contracts/players.html

Relationships:
Each of the data sources under consideration share the NBA player's name. Therefore, we are able to connect the different datasets by Player Name.

Next Steps:

- Approach:
  After initial review of the datasets, scope will be defined for volume (i.e. - Top 25 players versus Top 200), timeframe (i.e. – most recent five seasons), and define what stats contribute to "performance." Validate that the data sources meet project requirements. Review and clean data so that it is clear and formatted, such as standardizing player name and other data inputs. Configure datasets for primary key of Player Name and merge them to create a wholistic dataset of the information. Identify trends and insights for determining the market value of NBA players.
- Challenges:
  One of the key challenges of this project will be the cleaning and formatting of data from difference sources, specifically the primary key of Player's Name. Naming conventions across the datasets include: FirstName LastName, LastName FirstName, or the entire name in one field. In addition, salary amounts may differ by use of comma separators or abbreviations (i.e. – $4,000,000 versus $4M). There is also a concern of mis-matched or missing stats and how to handle them correctly. Project time will be invested into understanding these challenges clearly to determine a resolution in Python. In addition, the salary may reflect intangible variables not easily captured in stats (i.e. – marketing appeal, veteran status, or non-statistical contributions).

- Ethical Implications:
  All data used is from publicly available sources. Objectivity and fairness should be considered during analysis. There should be no personal labels attributed to players (i.e. – “not worth it”). This is an analytical exercise and will not contribute to the public scrutiny of NBA players. Identify proper processes and document the handling of missing data. Ensure conclusions are objective and can be attributed to the non-bias data.